IU02-Assessment Assignment

Module Name	Database Design and Implementation
Course Name	Foundation Diploma in Software Engineering
Assignment Title	MetroCity College Record Management Review

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Learner declaration
I certify that the work submitted for this assignment is my own and research sources are fully acknowledged. Student signature: star   Date: 6/28/2026

Contents

1. Project Background

MetroCity College is a growing higher education institution that offers diploma and degree programmes across multiple departments. Each semester, the college manages a large amount of information, including student profiles, course details, enrolment records, and exam results. Currently, most administrative records are maintained manually or through separate spreadsheets by different departments.

As the number of students, courses, and enrolments continues to increase, the current system has become difficult to manage and prone to inconsistencies. Repeated data, update errors, and slow searching can affect the accuracy and efficiency of record management.

2. Project Objective

The objective of this project is to develop a simple and beginner-friendly database design for MetroCity College to improve the management of student, course, enrolment, and exam score records.

Provide a simple database planning solution for the MetroCity College system by creating:

- A database content and requirement list
- A normalization designs up to Third Normal Form (3NF)
- An Entity-Relationship Diagram (ERD) plan
- A final relational schema with primary keys and foreign keys
- A short explanation of how the design supports college administration

3. Task 1 - Database Content Planning

The MetroCity College database should store the following information to support the management of students, courses, enrolments, and examination records.

1. Student Details

- Student ID
- Student Name
- Email Address
- Phone Number

2. Department Details

- Department ID
- Department Name
- Department Location

3. Course Details

- Course ID
- Course Name
- Department

5. Enrolment Details

- Enrolment ID
- Student
- Course
- Enrolment Date
- Enrolment Status

6. More Info Details

- Exam score, grade, and exam result details for each student
- Which student is enrolled in which course and semester
- Whether a course is active or inactive
- Staff actions such as registering students, managing courses, updating enrolments, and recording exam results

4. Task 2 - Normalization Design

1.Unnormalized Form (UNF)

Initially, all information may be stored in a single spreadsheet as shown below.

StudentID	StudentName	Department	CourseID	CourseName	Enorlment date	ExamName	ExamScore
ST001	Kaung Kaung	Computing	CS101, CS102	Database, Programming	01/06/2026, 02/06/2026	Midterm, Final	80, 85

Problems:

- Multiple values are stored in a single cell.
- Student, course, and exam information are repeated.
- Update, insertion, and deletion anomalies may occur.

2. First Normal Form (1NF)

In 1NF, each cell contains only one atomic value.

StudentID	StudentName	Department	CourseID	CourseName	Enrolment date	ExamName	ExamScore
ST001	Kaung Kaung	Computing	CS101	Database	01/06/2026	Midterm	80
ST001	Kaung Kaung	Computing	CS102	Programming	02/06/2026	Final	85

Improvement:

- Multiple values are removed from each cell.
- Each field contains only one value.

This table is now clearer, but department , semester and student information are still repeated. Therefore, the design must continue to Second Normal Form.

3. Second Normal Form (2NF)

In 2NF, data that depends only on part of the composite key is separated into different tables.

Student Table

StudentID	StudentName	Department
ST001	Kaung Kaung	Computing

Course Table

CourseID	CourseName
CS101	Database
CS102	Programming

Enrollment Table

EnrolmentID	StudentID	CourseID	EnrolmentDate
E001	ST001	CS101	01/06/2026
E002	ST001	CS102	02/06/2026

Exam Table

ExamID	StudentID	CourseID	ExamName	ExamScore
EX001	ST001	CS101	Midterm	80
EX002	ST001	CS102	Final	85

Improvement:

- Student and course information are no longer repeated.
- Partial dependencies are removed.

4. Third Normal Form (3NF)

Third Normal Form ensures that every non-key attribute depends only on the primary key of its own table. The final design adds the remaining useful attributes to the correct tables.

Student Table

StudentID (PK)	StudentName	StudentEmail	StudentPh-Number	DepartmentID (FK)
ST001	Kaung Kaung	Kaung4e@gmail.com	09-485745634	D001

Department Table

DepartmentID (PK)	DepartmentName	DepartmentLocation
D001	Computing	Building A, Floor 2

Course Table

CourseID	CourseName
CS101	Database
CS102	Programming

Enrollment Table

EnrolmentID	StudentID	CourseID	EnrolmentDate
E001	ST001	CS101	01/06/2026
E002	ST001	CS102	02/06/2026

Exam Table

ExamID	CourseID	ExamName
EX001	CS101	Midterm
EX002	CS102	Final

Exam Result Table

ResultID (PK)	StudentID (FK)	ExamID (FK)	ExamScore	ExamStatus
R001	ST001	EX001	80	Pass
R002	ST001	EX002	85	Pass

Improvement:

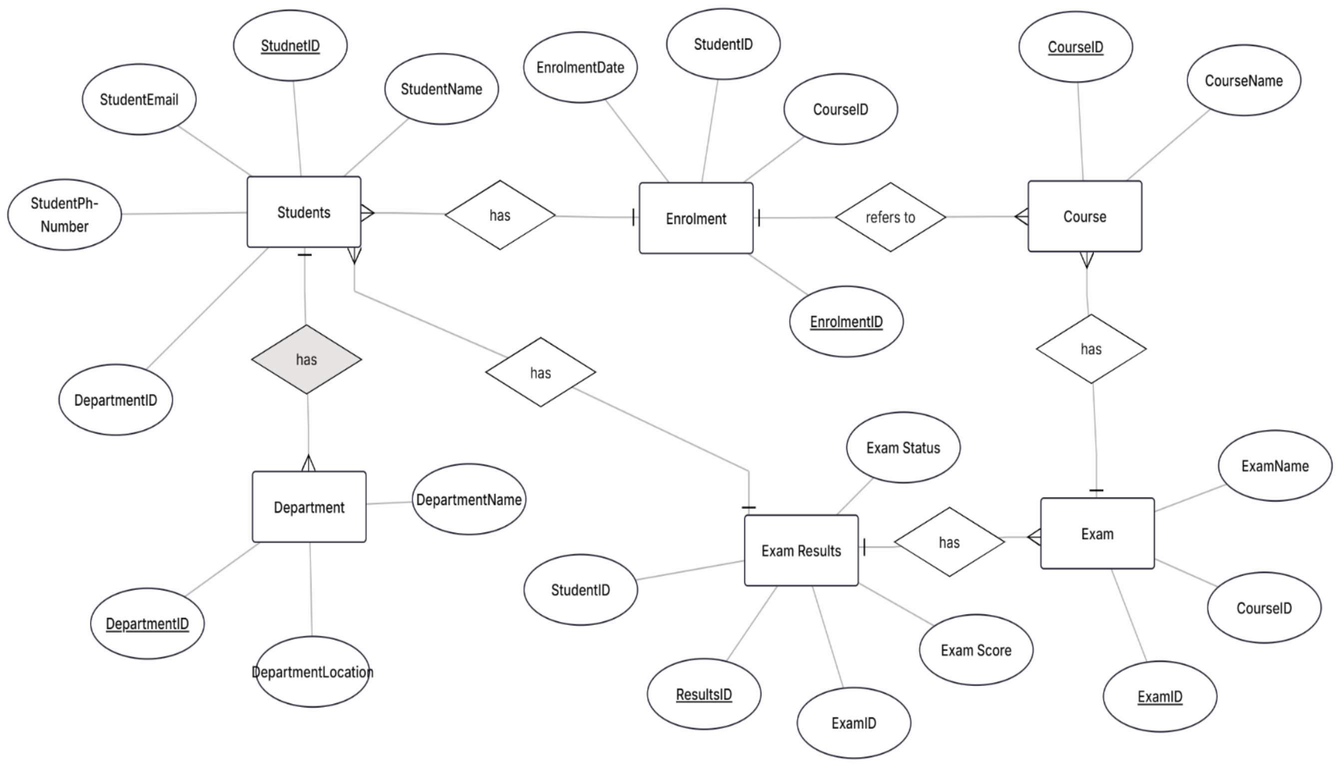
- Data redundancy is greatly reduced.
- Update, insertion, and deletion anomalies are minimized.
- Information is stored more efficiently and consistently.

Normalization reduces repeated data by storing information only once in separate tables. This minimizes update, insertion, and deletion anomalies, improves data consistency, and makes the database easier to manage.

5. Task 3 - ER Diagram Plan

The ER diagram is created from the final 3NF tables. The main entities are students, courses, enrolment , department , exam and exam results .

Simple ERD idea:

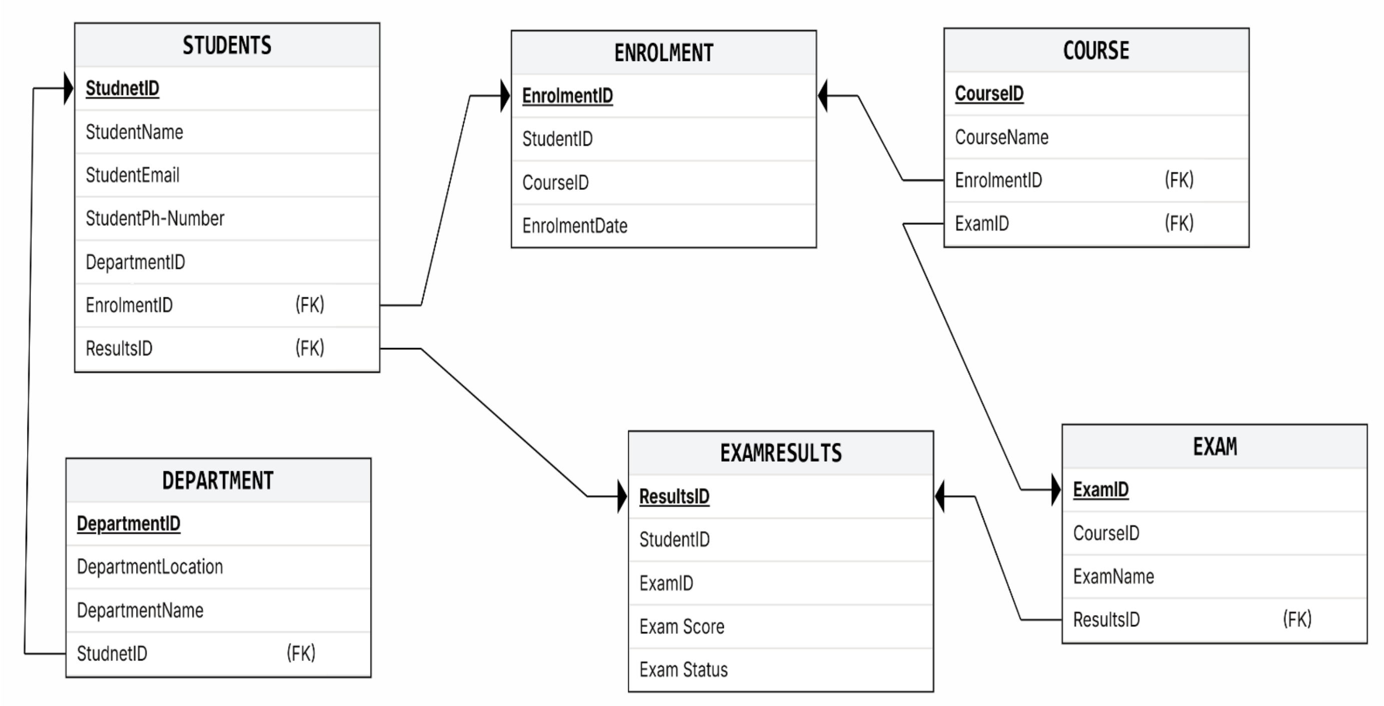


- Enrollment acts as the linking entity between Student and Course because it records each course enrollment made by a student.
- Exam Result acts as the linking entity between Student and Exam because it records the score obtained by each student in a particular exam.
- Department is related to Student because each student belongs to one department, while a department can have many students.
- Department is related to Course because each department can offer many courses, while each course belongs to one department.
- Exam is related to Course because each course can have multiple exams, while each exam belongs to one course.

6. Task 4 - Relational Schema and Design Explanation

Based on normalization and the ER diagram, the final relational schema is:

- Department (DepartmentID PK, DepartmentName, DepartmentLocation)
- Student (StudentID PK, StudentName, StudentEmail, StudentPh-Number, DepartmentID FK)
- Course (CourseID PK, CourseName)
- Enrollment (EnrollmentID PK, StudentID FK, CourseID FK, EnrollmentDate)
- Exam (ExamID PK, CourseID FK, ExamName)
- ExamResults (ResultsID PK, StudentID FK, ExamID FK, ExamScore, ExamStatus)



Foreign key rules:

- Student.DepartmentID references Department.DepartmentID
- Course.DepartmentID references Department.DepartmentID
- Enrollment.StudentID references Student.StudentID
- Enrollment.CourseID references Course.CourseID
- Exam.CourseID references Course.CourseID
- ExamResult.StudentID references Student.StudentID

- ExamResult.ExamID references Exam.ExamID

These foreign keys ensure that records are properly linked and that enrollment, exam, and result records cannot exist without valid student, course, and exam information.

Short Database Design Explanation

The database design helps MetroCity College organize student, course, enrollment, and examination information in a structured and efficient manner. By storing data in separate but related tables, repeated information is reduced and data consistency is improved. The use of primary keys and foreign keys ensures that records are accurately linked, making it easier to search, update, and manage information. Overall, the design supports reliable record management and prepares the college for future database implementation using SQL and a Database Management System (DBMS).